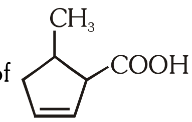
CARBOXYLIC ACIDS & ITS DERIVATIVES

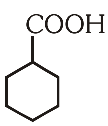
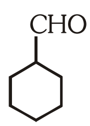
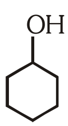
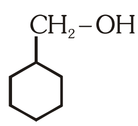
1. IUPAC name of  is

(1) 2-Methyl cyclopent-4-ene carboxylic acid
(2) 5-Methyl cyclopent-2-ene carboxylic acid
(3) 5-Methyl cyclopent-2-enoic acid
(4) 5-Methyl cyclopentene methanoic acid

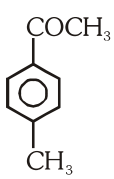
2. The carboxylic carbon of acid is less electrophilic than carbonyl carbon of Aldehyde/Ketone due to–
(1) more positive charge on carboxylic carbon
(2) less positive charge on carboxylic carbon because of resonance
(3) trigonal planar nature of carboxylic carbon
(4) tautomerism in carbonyl compound

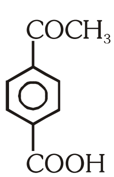
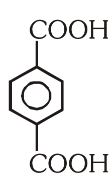
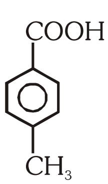
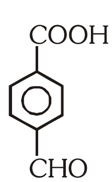
3.  $\xrightarrow[h\nu]{Br_2}$ A $\xrightarrow[\text{dry ether}]{Mg}$ B $\xrightarrow[(ii)H_3O^{\oplus}]{(i)CO_2}$ C

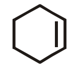
'C' is

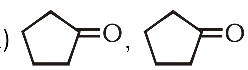
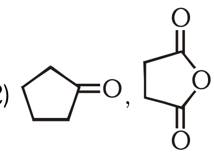
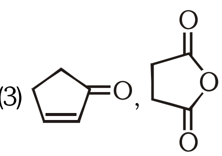
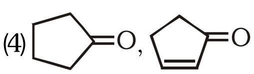
(1) 
(2) 
(3) 
(4) 

4. $CH_3-CH_2-CH_2-COOC_2H_5 \xrightarrow{NaOH}$ A 'A' is
(1) $CH_3-CH_2-CH_2-COOH + C_2H_5-OH$
(2) $CH_3-CH_2-CH_2-COONa$ + Et OH
(3) $CH_3-CH_2-CH_2-COOH + C_2H_5-O^-$
(4) $CH_3-CH_2-COO^- + C_2H_5O^-$

5.  $\xrightarrow[(ii)H_3O^{\oplus}]{(i)KMnO_4/^-OH/\Delta}$ A, 'A' is

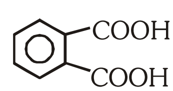
(1) 
(2) 
(3) 
(4) 

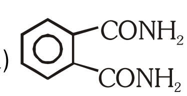
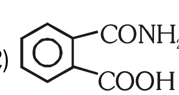
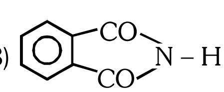
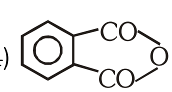
6.  $\xrightarrow[H^{\oplus}]{KMnO_4}$ A — $\xrightarrow[\Delta]{Ca(OH)_2}$ B ; $\xrightarrow{\Delta}$ C   B and C are

(1) 
(2) 
(3) 
(4) 

7. Correct acidic order is
(1) $CCl_3-COOH > F-CH_2-COOH > Cl-CH_2-COOH > Ph-COOH$
(2) $O_2N-CH_2-COOH > CF_3-COOH > Ph-COOH > CH_3-COOH$
(3) $F-CH_2-COOH > CCl_3-COOH > Cl-CH_2-COOH > Ph-COOH$
(4) $CF_3COOH > O_2N-CH_2-COOH > CH_3-COOH > Ph-COOH$

8. Esterification in acidic medium
(1) acid activates carbonyl carbon towards nucleophilic attack
(2) Acid deactivates carbonyl carbon
(3) is irreversible
(4) does not take place

9.  + $NH_3 \xrightarrow[\text{(strong)}]{\Delta}$ A 'A' is

(1) 
(2) 
(3) 
(4) 

10. $CH_3CH_2COOH + Cl_2 \xrightarrow{\text{Red P}} A \xrightarrow[\Delta]{NaOH/CaO} B \xrightarrow[\Delta]{Alc.KOH}$ C 'C' is
(1) $CH_3-CH(OH)-CH_3$
(2) $CH_3-CH=CH_2$
(3) $CH_3-CH_2-C(=O)-OR$
(4) $CH_2=CH_2$

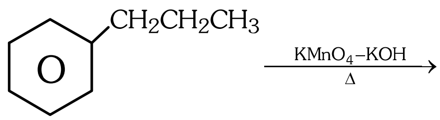
11 Carboxylic carbon is................than carbonyl carbon:

(1) less electrophilic

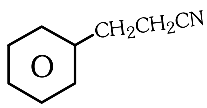
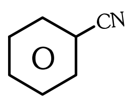
(2) more electrophilic

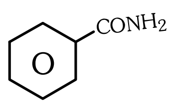
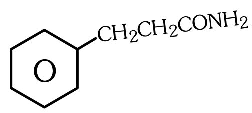
(3) less nucleophilic

(4) more nucleophilic

12 

$X \xrightarrow{H_3O^{\oplus}} Y \xrightarrow[(ii)\ \Delta]{(i) NH_3} Z$; Z is–

(1)  (2) 

(3)  (4) 

13 $RMgX + CO_2 \xrightarrow{\text{dry ether}} P \xrightarrow{H_3O^{\oplus}} Q$; Q can also be obtained by:

(1) hydrolysis of RCN

(2) hydrolysis of $RCONH_2$

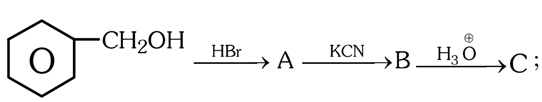
(3) oxidation of $R\text{–}CH_2OH$

(4) All

14 Hydrolysis of which of the following gives acid at fastest rate:

(1) $R-\overset{O}{\overset{\|}{C}}-OR$ (2) $R-\overset{O}{\overset{\|}{C}}-Cl$

(3) $R-\overset{O}{\overset{\|}{C}}-NH_2$ (4) $R-\overset{O}{\overset{\|}{C}}-O-\overset{O}{\overset{\|}{C}}-R$

15 

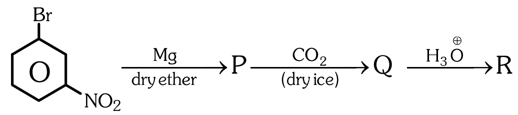
C is–

(1) Phenyl ethanoic acid

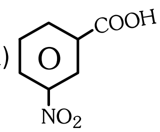
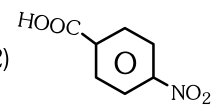
(2) Benzoic acid

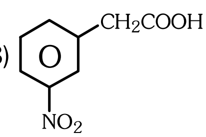
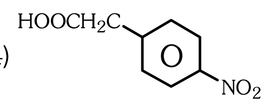
(3) Benzyl amine

(4) Aniline

16 

R is–

(1)  (2) 

(3)  (4) 

17 Which of the following statement is true:

(1) Most of the carboxylic acid exist as dimer in the vapour phase

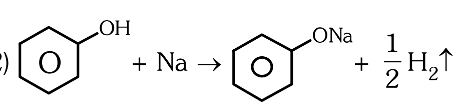
(2) Higher carboxylic acid are practically insoluble in water due to increased hydrophobic interaction

(3) Amides can also be obtained by hydrolysis of cyanides

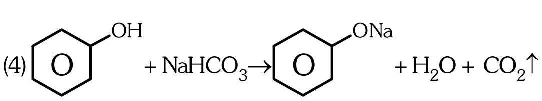
(4) All

18 Which of the following is incorrect:

(1) $R\text{–}COOH + Na \rightarrow R\text{–}COONa + \frac{1}{2}H_2\uparrow$

(2) 

(3) $R\text{–}COOH + NaHCO_3 \rightarrow R\text{–}COONa + H_2O + CO_2\uparrow$

(4) 

19 Which of the following can not be used to reduce ester

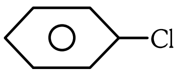
(1) $LiAlH_4 / H_3\overset{\oplus}{O}$

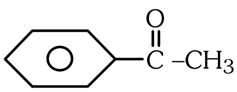
(2) DIBAL–H / H_2O

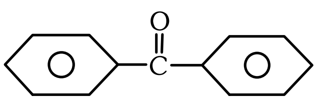
(3) $NaBH_4/C_2H_5OH$

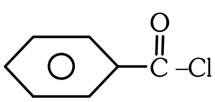
(4) $Na + C_2H_5OH$

20 $H_3C-COOH \xrightarrow[\Delta]{P_2O_5} A \xrightarrow{PCl_5} B \xrightarrow[AlCl_3]{C_6H_6} C$; C is–

(1) 

(2) 

(3) 

(4) 

21 Which of the following is not correctly matched

Compound	Use
(1) Ethanoic acid	Solvent
(2) Ester of benzoic acid	Perfumes
(3) Sodium benzoate	Preservatives
(4) Iodoform	Aneasthetic

22 In which of the following reaction product has acidic strength more than the initial compound used in reaction:

(1) Kolbe's electrolysis

(2) Hell-Volhard-Zelinsky reaction

(3) Sodalime decarboxylation

(4) Esterification

23 Which of the following is the correct order of effect of the following groups in increasing acidity order when α hydrogen of acetic acid is replaced by the following groups–

(1) $-F < -Cl < -Ph < -NO_2$

(2) $-Ph < -Cl < -F < -NO_2$

(3) $-Ph < -Cl < -NO_2 < -F$

(4) $-NO_2 < -F < -Cl < -Ph$

24 Identify the end product:

$$CH_3CH_2CH_2CHO \xrightarrow{\text{Ammonical } AgNO_3} X \xrightarrow{NaHCO_3} Y \xrightarrow[\Delta]{(NaOH+CaO)} Z$$

(1) $CH_3-CH_2CH_2-COOH$

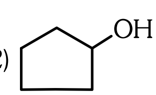
(2) $CH_3-CH_2-CH_2-CH_3$

(3) CH_3-CH_2-COOH

(4) $CH_3-CH_2-CH_3$

25 An organic compound contains 69.77% carbon, 11.63% hydrogen and rest of oxygen. The molecular mass of the compound is 86. It does not reduce tollen's reagent but forms an addition compound with sodium hydrogen sulphite and gives positive iodoform test. On vigrous oxidation it gives ethanoic acid and propanoic acid. The possible structure of the compound is:

(1) $CH_3-\underset{\displaystyle OH}{\underset{|}{CH}}-CH_2-CH=CH_2$

(2) 

(3) $CH_3-CH_2-\underset{\displaystyle O}{\underset{\|}{C}}-CH_2-CH_3$

(4) $CH_3-\underset{\displaystyle O}{\underset{\|}{C}}-CH_2-CH_2-CH_3$